

सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उदयम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

Ref: CTU/N/00/CMETS_NR/16

Date: 28-03-2023

As per distribution list

Subject: 16th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

Dear Sir/Ma'am,

Please find enclosed the minutes of the 16th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 28th February, 2023 (Tuesday) through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,

Kashish Bhambhani 28/03/23

(Kashish Bhambhani)
General Manager (CTU)

Distribution List:

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Minutes of 16th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 28.02.2023

The 16th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 28/02/2023. List of participants is enclosed at **Annexure-X**.

A. Confirmation on the minutes of the 15th CMETS-NR meeting

It was deliberated that the minutes of the 15th CMETS-NR meeting held on 31/01/2023 were issued vide letter dated 27/02/2023. However, it was informed that in the minutes circulated, for LTA application of M/s Tepsol Sun Sparkle Private Limited for 300 MW at Bhadla-III PS, the end date of LTA was inadvertently mentioned as 30/10/2048, however, it may be read as 30/09/2050. Other than above, no comments were received, accordingly, minutes were confirmed as circulated. The detailed agenda was discussed as below:

B. Application related matters in Northern Region

1. **Proposal for grant of Connectivity to the applications received from Renewable Energy Sources**

It was deliberated that the following 9 Nos. of Stage-I Connectivity applications were received in the month of Jan'23 and were agreed to be granted as per the details below:

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
1.	0112100033	Serentica Renewables India Private Limited	Barmer distt., Rajasthan	11.01.2023	31.12.2024/ 500	Generator (Solar)	Fatehgarh -IV PS	Common Pooling station of Serentica Renewables India Private Limited for 500 MW each Solar Power Project – Fatehgarh -IV PS (Section-2) 400 kV S/c (high capacity) line
2.	0112100044	Serentica Renewables India Private Limited	Barmer distt., Rajasthan	30.01.2023	31.12.2024/ 500	Generator (Solar)	Fatehgarh -IV PS	(Suitable to carry minimum 1000 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
3.	0112100038	Deshraj Solar Energy Private Limited	Bikaner distt., Rajasthan	23.01.2023	31.01.2025/ 400	Generator (Solar)	Bikaner- III PS	Deshraj Solar Energy Private Limited Solar Power Project – Bikaner- III PS 220 kV S/c (High Capacity) line (Suitable to carry minimum 400 MW at nominal voltage) Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.
4.	0112100039	Luceo Solar Private Limited	Barmer distt., Rajasthan	26.01.2023	31.12.2024/ 100	Generator (Solar)	Fatehgarh -IV PS	Common pooling station of M/s Luceo Solar Private Limited Solar power project of 200 MW & 100 MW each – Fatehgarh -IV (Section-2) PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
5.	0112100043	Helia Energy Park Private Limited	Barmer distt., Rajasthan	30.01.2023	31.12.2024/ 300	Renewable Power Park Developer (Solar)	Fatehgarh -IV PS	Helia Energy Park Private Limited Renewable Power Park Project – Fatehgarh-IV (Section-2) PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
6.	0112100041	Saimaa Solar Private Limited	Bikaner distt., Rajasthan	31.01.2023	15.01.2025/ 400	Generator (Solar)	Bikaner- III PS	Saimaa Solar Private Limited Solar Power Project – Bikaner- III PS 220 kV S/c (High Capacity) line (Suitable to carry minimum 400 MW at nominal voltage) Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.
7.	0112100046	Radiant Star Solar Park Private Limited	Barmer distt., Rajasthan	31.01.2023	31.12.2024/ 300	Renewable Power Park Developer (Solar)	Fatehgarh -IV PS	Radiant Star Solar Park Private Limited Renewable Power Park Project – Fatehgarh -IV PS (section-2) 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
8.	0112100047	Adani Solar Energy AP Three Limited	Jaisalmer distt., Rajasthan	31.01.2023	03.10.2023/ 150	Generator (Solar)	Ramgarh PS	Common Pooling station of M/s Adani Renewable Energy Holding Four Ltd. (500 MW) & M/s Adani Solar Energy AP Three Limited (150 MW) Solar Power Projects – Ramgarh PS 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage) Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
9.	0112100040	Resco Global Wind Services Private Limited	Jaisalmer distt., Rajasthan	31.01.2023	01.02.2025/ 500	Generator (wind)	Ramgarh PS	Resco Global Wind Services Private Limited Wind Power Project – Ramgarh PS 220 kV D/c line (Suitable to carry minimum 300 MW per circuit at nominal voltage) Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.

(i) Common Transmission system for Connectivity at Fatehgarh-IV PS (under ISTS)

- Establishment of 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) *
- or
- Establishment of 1x1500 MVA, 765/400 kV & 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS
- * with minimum capacity of 2100 MVA on each circuit at nominal voltage

(ii) Common Transmission system for Connectivity at Bikaner-III PS

- Establishment of 1x500MVA, 400/220 kV, Bikaner-III 765/400/220kV pooling station
- LILO of one Ckt of 400 kV Bikaner (PG)-Bikaner-II (Quad) D/c line at Bikaner-III PS or Bikaner-II – Bikaner-III 400 kV D/c line (Quad)

(iii) Common Transmission system for Connectivity at Ramgarh PS

- Establishment of 1x500 MVA, 400/220 kV & 1x1500 MVA 765/400 kV pooling station at Ramgarh PS
- Ramgarh PS - Bhadla-III PS 765 kV D/c line
- Bhadla-III PS -Sikar-II 765 kV D/c line
- It was mentioned that the grant of Stage-I Connectivity shall not create any right in favour of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the development of their generation project and associated transmission infrastructure as per format RCON-I-M (available on CTU website>>Stage-I / Stage-II Status Monitoring Portal) by 30th day of June and 31st December of each year.

- Further Stage-I Connectivity grantee is required to secure the Stage-II Connectivity for physical connectivity with the ISTS grid within 24 months. The Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I Grantee and their Application fees shall be forfeited.

Note: It was deliberated that in view of above Common Transmission system required for Connectivity at various ISTS pooling stations, 400/220 kV ICT shall not be required, if Connectivity is being granted at 400 kV. Same shall be suitably included in the respective St-I Intimations.

2. Stage-II Connectivity applications

It was deliberated that following 5 Nos. of Stage-II Connectivity applications were received in the month of Jan'23 and were decided to be granted as per the details below:

TABLE 2

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
1.	0312100017 (Enhancement)	Khaba Renewable Energy Private Limited (Refer Note-1)	Barmer distt., Rajasthan	02.01.2023	1200003132/300	Generator (Solar)	30.06.2026	50/ 30.06.2026	L&FC	Fatehgarh -IV PS#	➤ Applicant had withdrawn their St-II Connectivity application vide their email dated 28/02/2023. Applicant reiterated during the 16 th CMETS-NR meeting that it is technically & commercially challenging for them to share DTL with other St-II Connectivity Grantee, accordingly, they wish to withdraw their St-II Connectivity application (No. 0312100017) of 50 MW at Bikaner-II PS. Refer Note-1

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
2.	0212100034	Serentica Renewables India Private Limited	Barmer distt., Rajasthan	11.01.2023	0112100033/ 500	Generator (Solar)	31.12.2024	300/ 31.12.2024 Revised: 30.09.2025 (Interim) ^{\$}	L&FC	Fatehgarh -IV PS [#]	Common Pooling station of M/s Serentica Renewables India Private Limited for Solar Power Projects with Appl. Nos. 0212100034 (300 MW) & 0212100036 (300 MW) – Fatehgarh-IV PS (section-2) [#] 220 kV D/c line (Suitable to carry minimum 300 MW per circuit at nominal voltage) - Under the scope of applicant
3.	0212100036	Serentica Renewables India Private Limited	Barmer distt., Rajasthan	30.01.2023	0112100044/ 500	Generator (Solar)	31.12.2024	300/ 31.12.2024 Revised: 30.09.2025 (Interim) ^{\$}	L&FC	Fatehgarh -IV PS [#]	Common 220 kV line bay at Fatehgarh-IV PS (section-2) shall be developed under ISTS. Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
4.	0312100018 (Enhancement)	Luceo Solar Private Limited	Barmer distt., Rajasthan	26.01.2023	0112100039/ 100	Generator (Solar)	31.12.2024	100/ 31.12.2024 Revised: 30.09.2025 (Interim) ^{\$}	L&FC	Fatehgarh -IV PS [#]	Common Pooling station of M/s Luceo Solar Private Limited for Solar Power Projects with Appl. Nos. 0212100032 (200 MW) & 0312100018 (100 MW) – Fatehgarh-IV PS (section-2) 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) - Under the scope of applicant

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
											Common 220 kV line bay at Fatehgarh-IV PS (section-2) shall be developed under ISTS. Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity
5.	0212100038	Adani Solar Energy AP Three Limited	Jaisalmer distt., Rajasthan	31.01.2023	0112100047/ 150	Generator (Solar)	03.10.2023	150/ 03.10.2023 Revised: 31.12.2024 (Interim) ^{\$}	PPA with PSPCL, Punjab	Ramgarh PS ^{##}	Common Pooling station of M/s Adani Renewable Energy Holding Four Ltd. (500 MW) & M/s Adani Solar Energy AP Three Limited (150 MW) Solar Power Projects with Appl. Nos. 1200002432 (500 MW) & 0212100038 (150 MW) – Ramgarh PS 400 kV S/c line on D/c tower (Suitable to carry minimum 900 MW at nominal voltage) - Under the scope of applicant Common 400 kV line bay at Ramgarh PS shall be developed under ISTS. Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity

^{\$} Final Start date of Connectivity shall be confirmed after award of ISTS works

(i) Common Transmission system for Connectivity at Fatehgarh-IV PS (under ISTS)

- Establishment 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) *

or

- Establishment of 1x1500 MVA, 765/400 kV & 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
 - LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS
- * with minimum capacity of 2100 MVA on each circuit at nominal voltage*

(ii) Common Transmission system for Connectivity at Ramgarh PS

- Establishment of 1x1500 MVA 765/400 kV pooling station at Ramgarh PS
- Ramgarh PS - Bhadla-III PS 765 kV D/c line
- Bhadla-III PS -Sikar-II 765 kV D/c line

Note:1 It was deliberated that Fatehgarh-IV S/s (Section-I) was planned for about 2 GW potential which is to be evacuated through 400 kV D/c line (with capacity of 2100 MVA/ckt) to Fatehgarh-3 S/s and the system beyond Fatehgarh-3. Accordingly, about 2060 MW of power is proposed to be evacuated through this 400 kV D/c line (2100 MVA) in Phase-III system from Fatehgarh-IV (Section-1). For evacuation of power from Fatehgarh- IV S/s(Section-2), the transmission System for Solar Energy Zones in Rajasthan Phase-IV(Part-2) system is required. Phase-IV(Part-2) system is already approved in 14th CMETS NR held on 23.12.2022 & 61st NRPC meeting held on 26/12/2022 and being taken to NCT for further approval. Therefore, considering above technical limitations in section-1 (beyond 2060 MW), no further St-II Connectivity/enhancement can be permitted in Section-1.

Accordingly, St-II Connectivity (50 MW) grant to M/s Khaba was proposed in the agenda for 16th CMETS-NR meeting through sharing of dedicated transmission system of M/s Cannice Renewables Energy Private Limited (Appl. Nos. 0212100004-80 MW & 1671600589106-150 MW) by M/s Khaba Renewable at Fatehgarh-IV PS.

Subsequently, M/s Khaba vide their letter dated 28/02/2023 had informed that they wish to withdraw their St-II Connectivity enhancement application (50 MW), as it is technically & commercially challenging for them to share DTL with other St-II Connectivity Grantee. Accordingly, during the 16th CMETS-NR meeting, M/s Khaba reiterated the same on withdrawal of their St-II Connectivity Application of 50 MW, which was accepted.

Tentative Start date of Connectivity at Fatehgarh-IV PS (section-2) is Sep'25. Final Start date shall be given after award of ISTS work. Above is part of Rajasthan SEZ Ph-IV (Part-2) system, which is being taken up in NCT meeting for approval & schedule for the same may be considered tentatively as Sep'25.

Tentative Start date of Connectivity at Ramgarh- PS is Dec'24. Final Start date shall be given after award of ISTS work.

3. GNA transition not opted by M/s Adani Renewable Energy Holding Four Ltd

It was deliberated that M/s Adani vide letter dated 22.02.23 had now informed that based on approval received from SECI for change in location, M/s Adani would not be seeking transition under GNA for this 1500 MW solar manufacturing project at Ramgarh. M/s Adani also requested not to take up bays (220kV bays-2 nos. and 400kV bay-1 no.) as well as transmission corridor (capacity 1500 MW) for above 1500 MW project. It is to mention that M/s Adani vide letter dated 12.11.22 had earlier opted for GNA transition for above project.

It was deliberated that GNA transition process is yet to be completed and GNA regulations are not fully effective. GNA transition process is expected to be completed in next 3-4 months.

Further, it was intimated to M/s Adani that this above quantum of 1500 MW shall be reallocated to other applicants/grantees at Ramgarh/Bhadla-III PS etc as per the Regulations. M/s Adani agreed for the same and also stated that their 1 No. of 400 kV & 2 Nos. of 220 kV Connectivity bays at Ramgarh PS need not be taken up for implementation under Rajasthan SEZ Ph-III scheme.

Accordingly, it was agreed that as M/s Adani had confirmed their denial for GNA Transition and requested to not take up their associated 400 kV & 220 kV connectivity bays at Ramgarh PS, which are part of Phase-III, Part-C1 under Rajasthan SEZ Phase-III Transmission scheme (Scheme is under bidding presently), the 3 Nos. of bays (1 at 400 kV and 2 at 220 kV) shall not be taken up for bidding and scope of works shall be amended thereof by the BPC.

4. Proposal for grant of LTA to the applications to be processed from the month of Dec'2022**TABLE 3**

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA as per application/discussion	End date of LTA
1	SW9990894382-M029_D001_A007-1671216520154	Prerak Greentech Solar Private Limited	17.12.2022	Bhadla-III PS, Rajasthan	0212100003	Target (WR)-300 MW Target (NR)-100 MW	400 (Target)	31/01/2025 Revised: 31/12/26 \$ (Interim)	30/01/2055
2	SW9999482858-M029_D001_A007-1671556542236	Amplus Ages Private Limited	20.12.2022	Bikaner-II PS, Rajasthan	1200003624	Target (WR)-08 MW Target (ER)-14 MW Target (NR)-20 MW Target (SR)-20 MW	62 (Target)	31/10/2023	30/10/2048
3	SW9990897533-M029_D001_A007-1672319027575	Tepsol Sun Sparkle Private Limited	29.12.2022	Bhadla-III PS, Rajasthan	0212100030	Target (WR)-100 MW Target (NR)-100 MW Target (SR)-100 MW	300 (Target)	01/10/2025	30/09/2050
4	0412100009	Juniper Green Stellar Private Limited	30.12.2022	Fatehgarh-IV PS, Rajasthan	1200003827-100MW, 1200003910-100MW,	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2025	30/06/2050
5	0412100010	Juniper Green Stellar Private Limited	30.12.2022	Fatehgarh-IV PS, Rajasthan	1200003910-100MW, 1200003958-60MW, 0312100010-45MW	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2026	30/06/2051

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA as per application/discussion	End date of LTA
6	0412100012	Juniper Green Beta Private Limited	30.12.2022	Bhadla-III PS, Rajasthan	0212100027	Target (WR)-50 MW Target (NR)- 50 MW Target (SR)-50 MW	150 (Target)	30/06/2025	30/06/2050
7	0412100013	Juniper Green Beta Private Limited	30.12.2022	Bhadla-III PS, Rajasthan	0312100013-40MW, 1670400777119-70MW	Target (NR)	100 (Target)	31/12/2026 Revised: 31/12/26 \$(Interim)	31/12/2051

\$Final Start date of LTA shall be given on award of ISTS works

A. Regarding LTA Applications at Bhadla-III PS (Sl. No. 1,3,6 & 7): It was deliberated in the 15th & 16th NR-CMETS meeting held on 31/01/2023 & 28/02/2023 respectively that based on the system studies, the power transfer from Bhadla-III PS is envisaged with the Transmission system associated with REZ in Rajasthan under 20 GW Phase-III scheme. The transmission system has been approved in the 3rd NRPC(TP) meeting held on 19/02/2021 & 5th NCT meeting held on 25/08/2021 & 02/09/2021 respectively. It was also informed that establishment of 765/400/220 kV Bhadla-III PS along with onwards Corridors i.e. 765 kV Bhadla-III-Sikar-II-Khetri/Narela D/c is under bidding (Ph-III) with 18 months implementation schedule, whereas, 400 kV Bhadla-III-Bhadla (HVDC) 2xD/c & 6 GW Bhadla (HVDC)- Fatehpur HVDC system has been recently approved by MoP (with 42 months schedule). Considering the bidding as well as execution timeline, HVDC system is expected by Dec'26 tentatively.

Further, 765/400/220kV Ramgarh PS is also being connected with Bhadla-III PS. Therefore, corridors beyond Bhadla-III PS shall be utilized for transfer of power from RE Generators at Ramgarh/Bhadla-III. As per the above planned transmission system, about 2900 MW can be evacuated utilizing EHVAC corridors i.e. 765kV Bhadla-III-Sikar-II D/c corridor. Out of above, based on receipt of applications, 2600 MW LTA (with connectivity at Ramgarh PS) is already granted utilizing above margins. Therefore, the balance quantum of about 300 MW is available for power transfer beyond Bhadla-III PS utilizing EHV AC corridor. For Power transfer requirement beyond 2900 MW, as per the above planned system, HVDC system shall be utilized for power transfer from Bhadla-III PS onwards.

Now, as mentioned in section (3) above, M/s Adani Renewables vide their letter dated 22/02/2023 had informed that based on approval received from SECI for change in location, they would not be seeking transition under GNA for this 1500 MW solar manufacturing project

at Ramgarh. Further, M/s Adani had also requested not to take up bays (220kV bays-2 nos. and 400kV bay-1 no.) as well as transmission corridor (capacity 1500 MW) for above 1500 MW project.

It was also deliberated that GNA transition process is yet to be completed and GNA regulations are not fully effective. GNA transition process is expected to be completed in next 3-4 months. It was deliberated that CTU is in receipt of total 950 MW of LTA from 4 Nos. of applicants at Bhadla-III PS and system studies were carried out for grant of LTA at Bhadla-III PS in a comprehensive manner, considering all the LTA applications together.

Accordingly, at present LTAs shall be granted through identified Ph-III HVDC corridor (interim time schedule -Dec'26). However, upon completion of GNA transition process, based on vacation of transmission capacity(ies) & availability of margins (1800 MW including M/s Adani's above request for 1500 MW GNA not transitioned), transmission system required for above applications (950 MW) will be reviewed in order of priority and be revised suitably.

B. Regarding LTA Applications with drawl in Southern Region (Sl. No. 2, 3, 4, 5, 6): It was deliberated that during the 15th & 16th CMETS-SR meeting held on 30/01/23 & 28/02/2023, matter regarding grant of LTA for 3450 MW for drawl of power by beneficiaries in SR and injection at various nodes in NR, WR and ER was discussed. As total power transfer requirement is exceeding the expected enhancement in ATC with the additional under implementation inter-regional links, the detailed system studies shall be carried out with the Southern Region stake holders for identification of new Inter-Regional links as per power flow requirements. Further, it was mentioned that the margins available shall be reserved as per the priority of LTA applications in ISTS between NEW Grid and SR Grid for grant of LTA towards import of 3450 MW power in SR from different injection points in NEW Grid. Further, a Joint study meeting was also held among SR Constituents from 21st-22nd Feb'23 for evolving the required Inter-regional corridor or explore ATC/TTC margins. However, the allocation of available margin/evolution of new transmission corridor needs to be finalized among SR Constituents in SR-CMETS meeting.

Accordingly, for LTA applications at Sl. No. 2,3, 4, 5 & 6, any additional system requirement for drawl of power in SR shall be deliberated in the CMETS-SR meeting and shall be granted LTA after approval in SR-CMETS meeting regarding drawl in SR.

For application at Sl. No. 1: It was deliberated that Stage-II Connectivity (0212100003, 400 MW) was granted to M/s Prerak Greentech Solar Private Limited at Bhadla-III PS through 220 kV S/c (high capacity) line. 1 No. 220 kV bay at Bhadla-III PS is being implemented under ISTS. Applicant confirmed the same. Now, M/s Prerak Greentech has applied for LTA of 400 MW on target basis to WR (300 MW) & NR (100 MW).

As stated above in section 4(A), LTA at Bhadla-III PS shall be granted through identified Ph-III HVDC corridor, as part of Transmission system associated with REZ in Rajasthan under 20 GW Phase-III scheme with interim time schedule Dec'26 as of now. Accordingly, tentative Start date of the LTA for the same may be considered as Dec'26. Details of transmission system for present LTA is mentioned at **Annexure-I**. However as mentioned in section 4(A), upon completion of GNA transition process, based on vacation of transmission capacity(ies) & availability of margins (1800 MW including M/s Adani's above request for 1500 MW GNA not transitioned), transmission system required for above application will be reviewed.

However, it was also noted that the present LTA application is also for drawl of power in WR (target basis) which was discussed in the earlier 15th CMETS-WR meeting. Further, additional Scheme (WR-NR) to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose) is already under implementation (exp. May'24). Accordingly, it was agreed to grant LTA to M/s Prerak Greentech Solar Private Limited for 400 MW from Bhadla-III PS to NR & WR on target basis from 31/12/2026 (Interim) to 30/01/2055. Final Start date of LTA shall be given upon award of ISTS elements.

For application at Sl. No. 7: It was deliberated that Stage-II Connectivity for LTA Appl. No. at Sl. No. 7 was granted through Common pooling station of M/s Juniper Green Beta Private Limited Solar power project of 150 MW, 40 MW & 70 MW and 50 MW of M/s Frugal to Bhadla-III PS through 220 kV S/c line on D/c tower. Now, M/s Juniper Green Beta Private Limited has applied for LTA at Bhadla-III PS for 100 MW on Target region (NR).

As stated above in section 4(A), LTA at Bhadla-III PS shall be granted through identified Ph-III HVDC corridor, as part of Transmission system associated with REZ in Rajasthan under 20 GW Phase-III scheme with interim time schedule Dec'26 as of now. Accordingly, tentative Start date of the LTA for the same may be considered as Dec'26. Further applicant has also sought start date of LTA as 31st Dec'26, which matches likely schedule of HVDC system. Details of transmission system for present LTA is mentioned at **Annexure-II**. Accordingly, it was agreed to grant LTA to M/s Juniper Green Beta Private Limited for 100 MW from Bhadla-III PS to NR on target region basis from 31/12/2026 (Interim) to 31/12/2051. Final Start date of LTA shall be given upon award of ISTS elements.

5. Proposal for grant of LTA to the application received from RE Energy Sources in Jan'23

It was deliberated that the following 7 Nos. of LTA applications were received in the month of Jan'23. Details are as below:

TABLE 4

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA sought as per application	End date of LTA
1	0412100018	Azure Power India Private Limited	18.01.2023	Bhadla-II PS, Rajasthan	1200002403	TSECL (NER)	50 (PPA/PSA Without Noc)	Revised as per applicant request 20/07/2024@	Revised as per applicant request 19/07/2049
2	0412100019	Amp Energy Green Private Limited	24.01.2023	Fatehgarh-IV PS, Rajasthan	0312100005	MPSEZ Utilities Ltd- 110 MW (WR) CESC Limited-10 MW (ER)	120 (110 MW- Firm 10 MW- (PPA/PSA Without Noc))	11/01/2025@	11/01/2050
3	0412100020	XI Xergi Power Private Limited	25.01.2023	Fatehgarh-III PS, Rajasthan	1200002847	GRIDCO Limited (ER)	200 (PPA/PSA Without Noc)	10/07/2024@	09/07/2049
4	0412100021 (Change in Region)	Renew Surya Vihaan Private Limited	31.01.2023	Fatehgarh-III PS, Rajasthan	1200002695	UPCL, Uttarakhand (NR)	100 (PPA/PSA Without Noc)	30/09/2023@	29/09/2048
5	0412100022 (Change in Region)	Renew Surya Aayan Private Limited	31.01.2023	Fatehgarh-III PS, Rajasthan	1200002692	KSEB (SR)	300 (PPA/PSA Without Noc)	30/09/2023@	29/09/2048
6	0412100023 (Change in Region)	Renew Surya Roshni Private Limited	31.01.2023	Fatehgarh-III PS, Rajasthan	1200002628	WBSEDCL (ER): 100MW	100 (PPA/PSA Without Noc)	30/09/2023@	29/09/2048

@Start date of LTA is interim (pls refer section below); Final date shall be intimated after award of ISTS work

For application at Sl. No. 1: It was deliberated that Stage-II Connectivity (1200002403, 500 MW) was granted to M/s Azure Power India Private Limited at Bhadla-II PS through 400 kV S/c line. 1 No. 400 kV bay at Bhadla-II PS is being implemented under ISTS. Now, M/s Azure Power has applied for LTA of 50 MW on firm basis to TSECL (NER). Earlier M/s Azure was also granted LTA for 150 MW for above project. Applicant confirmed the same.

Based on the system studies, Rajasthan Ph-II transmission scheme is already agreed in 5th NRST meetings held on 13/09/2019. The required power transfer of 50 MW from Bhadla-II PS was envisaged under the said system. Details of transmission system for present LTA is mentioned at **Annexure-III**. It may be noted that as part of Rajasthan SEZ Phase-II Transmission schemes, Part-E [Bhadla-II PS-Sikar-II 765kV D/c line(2nd)] shall also be required for LTA. The same is under advance stage of bidding and its SPV yet to be transferred. Accordingly, its tentative commissioning schedule may be considered as Oct'24.

Further, as the LTA has been applied on firm basis, and for drawl of power by DISCOMS (TSECL) through STU network from ISTS grid, NOC from the concerned STU is required. CTU vide letter dated 18/01/2023 had also requested Tripura DISCOMS to issue the NoC, which is still awaited. Applicant was advised to take up the matter of NoC with TSECL (NER). It was stated that if NOC of concerned STU is not available or the NoC is not effective as on Start Date of LTA, the billing for payment of transmission charges shall be undertaken on the LTA grantee as per CERC Regulations, till such time NOC is made available and becomes effective. The NoC with conditions would be considered effective only upon fulfillment of such conditions.

It was deliberated that the subject LTA application was also discussed and agreed for grant in the NER-CMETS meeting held on 24/02/2023.

Accordingly, it was agreed to grant LTA to M/s Azure Power India Private Limited for 50 MW from Bhadla-II PS to NER (TSECL Discoms) on firm basis from 01/11/2024 (Interim) to 19/07/2049. Final Start date of LTA shall be given upon award of ISTS elements.

For application at Sl. No. 2: It was mentioned that the Stage-II Connectivity (0312100005, 120 MW) was granted to M/s Amp Energy Green Private Limited at Fatehgarh-IV PS through 220 kV S/c line. 1 No. 220 kV bay at Fatehgarh-IV PS is being implemented under ISTS. 220 kV line & bay is common for Application Nos. 1200003022 (130 MW), 0312100005 (120 MW), 0312100007 (50 MW) & 1200003719 (50 MW) of M/s Amp Energy & M/s Sprng respectively. Now, M/s Amp Energy has applied for LTA of 120 MW on firm basis to MPSEZ Utilities Ltd-110 MW (Gujarat) and CESC Limited (West Bengal) -10 MW (ER). Applicant confirmed the same.

Based on the system studies, the required power transfer of 120 MW from Fatehgarh-IV PS is envisaged with the Transmission system associated with REZ in Rajasthan under 20 GW Phase-III scheme. The transmission system has been approved in 3rd NRPC(TP)

meeting held on 19/02/2021 & 5th NCT meeting held on 25/08/2021 & 02/09/2021. The system is under bidding and tentative Start date of LTA at Fatehgarh-IV PS (Section-1) may be considered as Dec'24. The schedule to be confirmed upon award of the scheme. Details of transmission system for present LTA is mentioned at **Annexure-IV**.

Further, as the LTA has been applied on firm basis, and for drawl of power by utility/DISCOMS through STU (GETCO & WBSETCL) network from ISTS grid, NOC from the concerned STU is required. CTU vide letter dated 25/01/2023 had also requested to GETCO (Gujarat) & WBSETCL (West Bengal) to issue the NoC. NoC from GETCO has been received vide email dated 31/01/2023. Therefore, LTA for 110MW to MPSEZ Utilities Ltd shall be on firm basis. The remaining LTA shall be on PPA/PSA without NoC basis. Accordingly, applicant was requested to take up the matter of NoC with WBSETCL(ER). It was stated that if NOC of concerned STU is not available or the NoC is not effective as on Start Date of LTA, the billing for payment of transmission charges shall be undertaken on the LTA grantee as per CERC Regulations, till such time NOC is made available and becomes effective. The NoC with conditions would be considered effective only upon fulfillment of such conditions.

It was deliberated that the above LTA grant was also agreed in the WR-CMETS & ER-CMETS meeting held on 27/02/2023.

Accordingly, it was agreed to grant LTA to M/s Amp Energy Green Private Limited for 120 MW from Fatehgarh-IV PS to MPSEZ Utilities Ltd-110 MW (WR) and CESC Limited-10 MW (ER) on firm basis from 11/01/2025 (Interim) to 11/01/2050. Final Start date of LTA shall be given upon award of ISTS elements.

Further, it was also informed that LTA has been sought till 11/01/2050, whereas, PPA & PSA are valid till 10/01/2050. It was decided that the remaining period of LTA, i.e. 11/01/2050 shall be on Target region basis.

For application at Sl. No. 3: It was mentioned that the Stage-II Connectivity (1200002847, 400 MW) was granted to M/s XI Xergi Power Private Limited at Fatehgarh-III PS through 220 kV S/c line. 1 No. 220 kV bay at Fatehgarh-III PS is being implemented under ISTS. LTA for 200 MW is already granted to M/s Xergi for 200 MW (Application No. 0412100007). Now, M/s XI Xergi has applied for LTA of additional 200 MW on firm basis to GRIDCO Limited (ER). Applicant confirmed the same.

Based on the system studies, the required power transfer of 200 MW from Fatehgarh-III PS is envisaged with the Transmission system associated with SEZ in Rajasthan under 20 GW Phase-III scheme. The transmission system has been approved in 3rd NRPC(TP) meeting held on 19/02/2021 & 5th NCT meeting held on 25/08/2021 & 02/09/2021. The system is under bidding and tentative Start date of LTA at Fatehgarh-III PS may be considered as Dec'24. Details of transmission system for present LTA is mentioned at **Annexure-V**.

It was noted that the LTA has been applied on firm basis, and for further drawl of power by GRIDCO- ER discoms through STU network from ISTS, for which NOC from the concerned STU shall be required. CTU vide letter dated 25/01/2023 had also requested to GRIDCO (ER) to issue the NoC. Accordingly, applicant was requested to take up the matter of NoC with GRIDCO (ER). Regarding this, applicant informed that GRIDCO (ER) has issued NoC in this regard. The same was noted.

Further, it was deliberated that the subject LTA application was discussed and agreed in the ER-CMETS meeting held on 27/02/2023. Accordingly, it was agreed to grant LTA to M/s XI Xergi Power Private Limited for 200 MW from Fatehgarh-III PS to GRIDCO- ER (200 MW) from 31/12/2024 (Interim) to 09/07/2049. Final Start date of LTA shall be given upon award of ISTS elements.

For application at Sl. No. 4: It was informed that Stage-II Connectivity (1200002695, 100 MW) was granted to M/s Renew Surya Vihaan Private Limited at Fatehgarh-III PS through 220 kV S/c line. 1 No. of 220 kV bay at Fatehgarh-III PS is being implemented under ISTS. Further, it was noted that M/s Renew Surya Vihaan Private Limited (RSVPL) has been granted 100 MW LTA vide CTU letter dated 27/10/2020 on Target (ER) basis at Fatehgarh-III PS with following details:

A. Transmission system for present LTA at Fatehgarh-III PS

- 1) Establishment of 2x500 MVA (3rd & 4th) 400/220kV ICT at Fatehgarh-III Pooling station
- 2) Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line
- 3) Jaisalmer (RVPN) S/s - Fatehgarh-III PS 400 kV (Twin HTLS) D/c line

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

- 1) Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
- 2) Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
- 3) 1x240 MVA Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
- 4) Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVA at 400kV level & 2x330 MVA bus reactors at 765kV level at Sikar -II
- 5) Sikar-II – Neemrana 400kV D/c line (Twin HTLS)
- 6) Bhadla-II PS – Sikar-II 765kV D/c line
- 7) 1x330 MVA Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
- 8) 1x240 MVA Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
- 9) Sikar-II – Aligarh 765kV D/c line

10) 1x330 MVar switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line

It was mentioned that LTA for 100 MW was granted with the above-mentioned transmission system for Target (ER). The present application is to change 100 MW LTA from Target (ER) to UPCL, NR on firm basis with start date of LTA sought as 30/09/2023. Further, the applicant has mentioned that the date of relinquishment of LTA in Target (ER) may be considered as 30/09/2023. As the effective date of start of LTA in the changed region is the same as date of relinquishment in original region and the change in region is sought for entire capacity relinquished, thereof, no relinquishment charges are leviable in line with CERC order in Petition 92/MP/2015.

Further, based on the system studies, Rajasthan SEZ Phase-II Transmission schemes is already agreed in 5th NRSCT meeting held on 13/09/2019. The required power transfer of 100 MW from Fatehgarh-III PS was envisaged under the said system. It was mentioned that part of transmission scheme under Rajasthan SEZ Phase-II (8.1 GW), mentioned above is under implementation and is envisaged to be implemented completely tentatively by Feb'24. Details of transmission system for present LTA is mentioned at **Annexure-VI**.

Further, it was noted that the LTA has been applied on firm basis, and for further drawl of power by NR discoms through STU network from ISTS, for which NOC from the concerned STU shall be required. CTU vide letter dated 07/02/2023 has also requested to discoms to issue the NoC, which is still awaited. Accordingly, applicant was requested to take up the matter of NoC with UPCL (NR). It was stated that if NOC of concerned STU is not available or the NoC is not effective as on Start Date of LTA, the billing for payment of transmission charges shall be undertaken on the LTA grantee as per CERC Regulations, till such time NOC is made available and becomes effective. The NoC with conditions would be considered effective only upon fulfillment of such conditions.

Accordingly, it is was agreed to grant LTA to M/s RSVPL for 100 MW to UPCL, Uttarakhand (NR) from 01/03/2024(Interim) to 29/09/2048 with the above-mentioned transmission system for transfer of power. Final Start date of LTA shall be given upon award of ISTS elements.

For application at Sl. No. 5: It was informed that the Stage-II Connectivity (1200002692, 300 MW) was granted to M/s Renew Surya Aayan Private Limited at Fatehgarh-III PS through 220 kV S/c line. 1 No. 220 kV bay at Fatehgarh-III PS is being implemented under ISTS. Now, M/s Renew Surya Aayan had applied for LTA of 300 MW on firm basis to KSEB (SR). Further, it was noted that M/s Renew Surya Aayan Private Limited (RSAPL) has been granted 300 MW LTA vide CTU letter dated 27/10/2020 on Target (ER) basis at Fatehgarh-III PS with following details:

A. Transmission system for present LTA at Fatehgarh-III PS

- 1) Establishment of 2x500 MVA (3rd & 4th) 400/220kV ICT at Fatehgarh-III Pooling station
- 2) Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line

3) Jaisalmer (RVPN) S/s - Fatehgarh-III PS 400 kV (Twin HTLS) D/c line

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

- 1) Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
- 2) Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
- 3) 1x240 MVA Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
- 4) Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVA at 400kV level & 2x330 MVA bus reactors at 765kV level at Sikar -II
- 5) Sikar-II – Neemrana 400kV D/c line (Twin HTLS)
- 6) Bhadla-II PS – Sikar-II 765kV D/c line
- 7) 1x330 MVA Switchable line reactor for each circuit at Sikar-II end of Bhadla-II – Sikar-II 765kV D/c line
- 8) 1x240 MVA Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
- 9) Sikar-II – Aligarh 765kV D/c line
- 10) 1x330 MVA switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line

It was mentioned that LTA for 300 MW was granted with the above-mentioned transmission system with 300 MW for Target (ER). The present application is to change 300 MW LTA from Target (ER) to KSEB, SR on firm basis with start date of LTA as 30/09/2023. Further, the applicant had mentioned that the date of relinquishment of LTA in Target (ER) may be considered as 30/09/2023. As the effective date of start of LTA in the changed region is the same as date of relinquishment in original region and the change in region is sought for entire capacity relinquished, thereof, no relinquishment charges are leviable in line with CERC order in Petition 92/MP/2015.

Based on the system studies, Rajasthan SEZ Phase-II Transmission schemes is already agreed in 5th NRSCT meeting held on 13/09/2019. The required power transfer of 300 MW from Fatehgarh-III PS was envisaged under the said system. It may be mentioned that part of transmission scheme under Rajasthan SEZ Phase-II (8.1 GW), mentioned above is under implementation and is envisaged to be implemented tentatively by Feb'24. Details of transmission system for present LTA is mentioned at **Annexure-VII**.

Further, as mentioned above that for drawl of power in SR, any additional system requirement or availability of ATC/TTC margin for drawl of power in SR shall be deliberated in the ensuing CMETS-SR meeting. Accordingly, the subject LTA grant shall be discussed again in NR-CMETS meeting after the consent of Southern region constituents in next SR-CMETS.

For application at Sl. No. 6: It was mentioned that the Stage-II Connectivity (1200002628, 400 MW) was granted to M/s Renew Surya Roshni Private Limited at Fatehgarh-III PS through 220 kV S/c line. 1 No. 220 kV bay at Fatehgarh-III PS is being implemented under ISTS. Now, M/s Renew Surya Roshni has applied for LTA of 100 MW on firm basis to WBSEDCL (ER) (100MW). Further, it mentioned that M/s Renew Surya Roshni Private Limited (RSRPL) has been granted 400 MW LTA vide CTU letter dated 30/07/2021 on Target (ER) -100 MW, Target (WR)-100 MW & Target (NR)-200 MW basis at Fatehgarh-III PS with following details:

A. Transmission system for present LTA at Fatehgarh-III PS

1. Augmentation of 1x500 MVA (4th) 400/220kV ICT at Fatehgarh-III Pooling station
2. Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line
3. Jaisalmer (RVPN) S/s - Fatehgarh-III PS 400 kV (Twin HTLS) D/c line

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

1. Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
2. Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
3. 1x240 MVar Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
4. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVar at 400kV level & 2x330 MVar bus reactors at 765kV level at Sikar -II
5. Bhadla-II PS – Sikar-II 765kV D/c line
6. Sikar-II – Neemrana 400kV D/c line (Twin HTLS)*
7. 1x330 MVar Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
8. 1x240 MVar Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
9. Sikar-II – Aligarh 765kV D/c line
10. 1x330 MVar switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line
11. Establishment of 765/400 kV, 3X1500 MVA GIS substation at Narela with 765 kV (2x330 MVar) bus reactor and 400kV (1x125 MVar) bus reactor.
12. Khetri – Narela 765 kV D/c line
13. LILO of 765 kV Meerut- Bhiwani S/c line at Narela
14. 1x330 MVar Switchable line reactor for each circuit at Narela end of Khetri – Narela 765kV D/c line
15. Removal of LILO of Bawana – Mandola 400kV D/c (Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharani Bagh – Narela 400kV D/c(Quad) and Maharani Bagh-Gopalpur-Narela 400 kV D/c (Quad) lines.

16. 2 no of line bays at Narela each for Maharani Bagh – Narela 400 kV D/c (Quad) and Maharani Bagh –Gopalpur-Narela 400 kV D/c (Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s.

17. STATCOM:

Fatehgarh – II S/s: STATCOM: ± 600 MVar, 4x125 MVAR MSC, 2x125 MVar MSR

Bhadla – II S/s: STATCOM: ± 600 MVar, 4x125 MVAR MSC, 2x125 MVar MSR

*with minimum capacity of 2100 MVA on each circuit at nominal voltage

It was deliberated that M/s Renew had also requested for firming up for its other target region LTA to Northern Railway [NR](100 MW), IPCL [ER](100 MW), UPCL [NR] (UK)-100 MW which will be taken up separately.

Now, it was mentioned that the applicant wants to relinquish 60 MW LTA from NR and 40 MW LTA from WR and applied 100 MW LTA with WBSEDCL (ER) as drawee entity. Further, the applicant had mentioned that the date of relinquishment of LTA in Target (ER) may be considered as 30/09/2023. As the effective date of start of LTA in the changed region is the same as date of relinquishment in original region and the change in region is sought for entire capacity relinquished, thereof, no relinquishment charges are leviable in line with CERC order in Petition 92/MP/2015.

Based on the system studies, Rajasthan SEZ Phase-II Transmission schemes is already agreed in 5th NRSCT meeting held on 13/09/2019. The required power transfer of 100 MW from Fatehgarh-III PS was envisaged under the said system. It was mentioned that part of transmission scheme under Rajasthan SEZ Phase-II (8.1 GW), mentioned above is under implementation and is envisaged to be implemented completely tentatively by Feb'24. Details of transmission system for present LTA is mentioned at **Annexure-VIII**.

Further, it was deliberated that the LTA has been applied on firm basis and for further drawl of power by WBSEDCL- ER discoms through STU network from ISTS, NOC from the concerned STU shall be required. CTU vide letter dated 07/02/2023 has also requested to discoms to issue the NoC, which is still awaited. Accordingly, applicant was requested to take up the matter of NoC with WBSEDCL (ER). It was stated that if NOC of concerned STU is not available or the NoC is not effective as on Start Date of LTA, the billing for payment of transmission charges shall be undertaken on the LTA grantee as per CERC Regulations, till such time NOC is made available and becomes effective. The NoC with conditions would be considered effective only upon fulfillment of such conditions. Also, subject LTA grant was discussed & agreed in ER-CMETS meeting held on 27/02/2023.

Accordingly, it was deliberated that the start date of LTA shall be 01/03/2024. Keeping above in view, it was to grant LTA to M/s RSRPL for 100 MW to WBSEDCL: ER from 01/03/2024(Interim) to 29/09/2048 with the above-mentioned transmission system for transfer of power.

6. Proposal for grant of LTA to applications received in other regions for Injection/drawl in Northern region

It was deliberated that LTA applications were received in Western region for drawl in NR:

TABLE 5

Sl. No.	Application ID	Name of the Applicant	Submission Date	Quantum of LTA	Start Date of LTA	End Date of LTA	Injection Point	Drawl Point
1.	0430700014	Sarjan Realities Private Limited	12.01.2023	1100 (Target)	30/04/2025	29/04/2050	SRPL, Khavda, Kutch, Gujarat	NR: 400 MW SR: 700 MW
2.	0430700013	Sarjan Realities Private Limited	12.01.2023	1250 (Target)	30/04/2025	29/04/2050	SRPL, Khavda, Kutch, Gujarat	NR: 500 MW WR: 500 MW SR: 250 MW
3.	0430700011	Sarjan Realities Private Limited	12.01.2023	1150 (Target)	30/04/2025	29/04/2050	SRPL, Khavda, Kutch, Gujarat	ER: 600 MW NR: 550 MW
4.	0431400004	Serentica Renewables India 4 Private Limited	19.01.2023	100 (Firm)	31/12/2024	30/12/2049	Kallam PS, Maharashtra (WR)	Hindustan Zinc Limited, Kankroli S/s, Rajasthan (NR)
5.	0412100015	Serentica Renewables India 4 Private Limited	19.01.2023	100 (Firm)	31/03/2025	30/03/2050	Kallam PS, Maharashtra (WR)	Hindustan Zinc Limited, Kankroli S/s, Rajasthan (NR)

It was noted that the above LTA (1650 MW) involves power transfer from WR to NR and no constraints are envisaged in above transfer of power. The detailed transmission system for injection of power in WR was also discussed in the 16th CMETS-WR meeting held on 27/02/2023.

Further, reg. applications at Sl. no 1, 2 for SR drawl, it may be noted that the margin in ISTS for import of power by SR beneficiaries from NEW Grid of about 20460 MW has already been allocated under different LTA / MTOA granted / agreed for grant with existing / under implementation transmission system. As the total power transfer requirements are exceeding the expected enhancement in ATC with the additional under implementation inter-regional links, the detailed transmission system studies shall be carried along with the

Southern Region stake holders for identification of new Inter-Regional links as per power flow requirements. Further, the margins available are reserved as per the priority of LTA applications in ISTS between NEW Grid and SR Grid for grant of LTA towards import of 3450 MW power in SR from different injection points in NEW Grid. Any additional system requirement for drawl of power in SR shall be deliberated in the ensuing CMETS-SR meeting.

Further, regarding applications at Sl. no 4 & 5 with drawl at Kankroli (NR), detailed studies need to be carried out.

Accordingly, above LTA grant for drawl in SR shall be subject to consent all the constituents in SR-CMETS meeting and drawl at Kankroli (NR) subject to consent all the constituents in next NR-CMETS meeting.

7. **MTOA for Power Transfer from NR to SR**

It was stated that two no. of MTOA applications have been received in the month of Jan'23. Details of the same is given below:

TABLE 6

Sl. No.	Application ID	Name of the Applicant	Submission Date	Drawl Region	Quantum of MTOA (MW)	Start Date of MTOA	End Date of MTOA	Injection Point	Entity/Drawl Point
1	0552400013 (Original Application)	Southern Power Distribution Company of Telangana Limited	30.01.23	SR	769.01	01/07/2023	31/03/2024	Fatehgarh II PS : 169.33MW (105.83 +63.5) Bhadla II PS : 599.68 MW (176.38+211.65+211.65)	TSSPDCL, Telangana (SR)
2	0552400014 (Original Application)	Northern Power Distribution Company of Telangana Limited	30.01.23	SR	321	01/07/2023	31/03/2024	Fatehgarh II PS : 70.67MW (44.17 +26.5) Bhadla II PS : 250.33 MW (73.63+88.35+88.35)	TSNPDCL, Telangana (SR)

- 1) **Sl. No. 1: MTOA application of Southern Power Distribution Company of Telangana Limited for 769.01MW of power transfer from its solar power project to be connected/to be connected at Fatehgarh-II PS (169.33MW) and Bhadla-II (599.68MW) in Rajasthan (NR) to TSSPDCL, Telangana (SR)**

M/s Southern Power Distribution Company of Telangana Limited has applied for transfer of power from solar power project(s) at Fatehgarh-II PS (169.33MW) and Bhadla-II PS (599.68MW) in Rajasthan (NR) to TSSPDCL, Telangana (SR) through Medium Term Open Access from 01/07/2023 to 31/03/2024.

M/s NTPC was granted 550MW (300+250) Stage-II Connectivity through 400kV S/c line and 300MW Stage-II Connectivity through 220kV S/c line at 765/400/220kV Bhadla-II substation with 400kV bay and 220kV bay respectively at Bhadla-II in ISTS scope. Further M/s NTPC was also granted 240MW (150+90) Stage-II Connectivity through 220kV S/c line at 765/400/220kV Fatehgarh-II substation with 220kV bay at Fatehgarh-II PS in developer's (M/s NTPC) scope.

Details of LTA commensurate to MTOA of M/s TSSPDCL for 769.01MW [Fatehgarh II PS: 169.33MW (105.83 +63.5) and Bhadla II PS : 599.68 MW (176.38+211.65+211.65)] is as under:

TABLE 7

S.No	LTA Applicant and Capacity	Commensurate Connectivity application and its capacity	CTU LTA intimation date	LTA effective date	Transmission system for LTA	Timeline for LTA effective
1	M/s TSSPDCL – 176.375MW	M/s NTPC – 550MW (250+300) at 400kV level of Bhadla-II	28.01.21 18.08.21	29.12.21 or system whichever is later	Part of Transmission System for Rajasthan SEZ Phase-II (8.1GW) +Ph-I ICT (765/400kV)	LTA likely to be operationalized with commissioning of part of Rajasthan SEZ Phase-II Part-D System (Dec'23)
2	M/s TSSPDCL – 211.65MW		20.05.21 18.08.21	27.07.22 or system whichever is later		LTA likely to be operationalized with commissioning of part of Rajasthan SEZ Phase-II Part-D System (Dec'23)

3	M/s TSSPDCL – 211.65MW	M/s NTPC - 300MW at 220kV level of Bhadla-II	20.05.21 18.08.21	28.05.22 or system whichever is later	Part of Transmission System for Rajasthan SEZ Phase-II (8.1GW) + Ph-I ICT (765/400kV & 400/220kV)	
4	M/s TSSPDCL – 105.83MW	M/s NTPC - 240MW (150+90) at 220kV level of Fatehgarh-II	28.01.21 18.08.21	29.12.21 or system whichever is later	Part of Transmission System for Rajasthan SEZ Phase-II (8.1GW)	LTA likely to be operationalized upon commissioning of part of Rajasthan SEZ Phase-II, Prt-B System (01.03.2024)
4	M/s TSSPDCL – 63.5MW		30.07.21	19.02.22 or system whichever is later		

Sl. No. 2: MTOA application of Northern Power Distribution Company of Telangana Limited for 321MW of power transfer from its solar power project connected/to be connected at Fatehgarh-II PS (70.67MW) and Bhadla-II (250.33MW) in Rajasthan (NR) to TSNPDCL, Telangana (SR)

M/s Northern Power Distribution Company of Telangana Limited has applied for transfer of power from solar power project at Fatehgarh-II PS (70.67MW) and Bhadla-II (250.33MW) in Rajasthan (NR) to TSNPDCL, Telangana (SR) through Medium Term Open Access from 01/07/2023 to 31/03/2024.

M/s NTPC was granted 550MW (300+250) Stage-II Connectivity through 400kV S/c line and 300MW Stage-II Connectivity through 220kV S/c line at 765/400/220kV Bhadla-II substation, being implemented by M/s NTPC with 400kV bay and 220kV bay at Bhadla-II in ISTS scope. Further M/s NTPC was also granted 240MW (150+90) Stage-II Connectivity through 220kV S/c line respectively at 765/400/220kV Fatehgarh-II substation, being implemented by M/s NTPC with 220kV bay at Fatehgarh-II in developer's (M/s NTPC) scope.

Details of LTA commensurate to MTOA of M/s TSNPDCL – 321MW [Fatehgarh II PS : 70.67MW (44.17 +26.5) and Bhadla II PS : 250.33 MW (73.63+88.35+88.35)] is as under

TABLE 8

S.No	LTA Applicant and Capacity	Commensurate Connectivity application and its capacity	CTU LTA intimation date	LTA effective date	Transmission system for LTA	Timeline for transmission system
1	M/s TSNPDCL – 73.625MW	M/s NTPC – 550MW (250+300) at 400kV level of Bhadla-II	20.05.21 18.08.21	28.05.22 or system whichever is later	Part of Transmission System for Solar Energy Zones in Rajasthan (8.1GW) under Phase-II++Ph-I ICT (765/400kV)	LTA to be operationalized with commissioning of part of Rajasthan SEZ Phase-II Part-D System (Dec'23)
2	M/s TSNPDCL – 88.35MW		20.05.21 18.08.21	27.07.22 or system whichever is later		
3	M/s TSNPDCL – 88.35MW	M/s NTPC - 300MW at 220kV level of Bhadla-II	20.05.21 18.08.21	28.05.22 or system whichever is later		
4	M/s TSNPDCL – 44.17MW	M/s NTPC - 240MW (150+90) at 220kV level of Fatehgarh-II	20.05.21 18.08.21	28.05.22 or system whichever is later	Part of Transmission System for Rajasthan SEZ Phase-II (8.1GW) +	LTA likely to be operationalized upon commissioning of

5	M/s TSNPDCL – 26.5MW		18.08.21	19.02.22 or system whichever is later	+ Ph-I ICT (765/400kV & 400/220kV)	part of Rajasthan SEZ Phase-II, Part-B System (01.03.2024)
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As per the information available, the solar project is being developed by M/s NTPC and connected at Bhadla-II Substation (Total capacity : 850MW, Commissioned : 600MW) and Fatehgarh-II Substation (Total capacity : 240MW, Commissioned : 240MW).

System studies and results for MTOA

It was deliberated that System study has been carried out for transfer of 1090.01 MW (769.01+321) power from M/s NTPC solar generation project (1090MW) for 850MW injection at 765/400/220kV Bhadla-II substation [400kV (300MW +250 MW) and 220kV (300MW)] and 240MW (150+90) injection at 765/400/220kV Fatehgarh-II S/s at 220kV level, which is however 0.01MW additional than total Connectivity/LTA quantum.

As per CERC Connectivity Regulations/Detailed Procedure, Medium-term Open Access shall be granted if the resultant power flow can be accommodated in the existing transmission system or the transmission system under execution expected to be commissioned within next 6 calendar months. In view of the above, studies were carried out in Solar maximized scenario with high RE (Solar) generation in Rajasthan considering following Parts/Tr. elements of Ph-I/II system

- Augmentation of 1x500 MVA, 400/220kV ICT (4th) at Bhadla-II PS (220kV Section-I): scheduled to be commissioned by 31.03.23
- Augmentation of 1x500 MVA, 400/220kV ICT (5th) at Bhadla-II PS (220kV Section-I): scheduled to be commissioned by 30.04.23
- Phase-II Part-A & associated transmission system: scheduled to be commissioned from 31.07.2023 to 30.09.2023
- Phase-II, Part-F comprising Bikaner-II & associated transmission system: scheduled to be commissioned by 30.04.23
- Phase-II, Part F1: scheduled to be commissioned by 31.03.23 (commissioning matching with Phase-II Part F)

From the table above, it is emerged that for operationlization of LTA commensurate to above MTOA applications, many elements of Phase-II transmission system i.e. Part-B,C,D and G is required with implementation schedule progressively from Nov'23. The same would not be available in MTOA timeframe (01.07.23), hence not considered in the MTOA studies

It was stated that with above MTOA capacity, loading of 400kV Bhinmal - Zerda is beyond thermal limits i.e. 904MW (against 874MVA). The loading will further aggravate to 1007 MW in N-1 contingency (outage of 400kV Kankroli-Zerda line). In addition to above, Loading of 400kV Bhadla(PG)- Bhadla(RVPN) D/c line is also very high in N-1 Contingency (about 1618MW). The loading will be further aggravated (>2000MW in N-1 contingency) in case of delay in Phase-II, Part-A transmission scheme (Sch. 31.07.23).

NRLDC/GRID-INDIA stated that with growing RE capacity installation, oscillations are being faced in western Rajasthan in Bhadla-II & Fatehgarh-II complexes due to various reasons including non-commissioning of Phase-II transmission system. Considering above MTOA should not be granted in absence of additional transmission system (Phase-II).

It was deliberated that the requested MTOA with effect from 01/07/2023 cannot be accommodated due to above transmission constraints considering the commissioning of the identified transmission system in next six months i.e. part of Transmission System Rajasthan Ph-II SEZ (8.1GW) (Part-A/F/F1) which is expected to be commissioned progressively from Apr'23. GRID-INDIA also agreed for the same.

Considering above it was agreed that above that two nos. of MTOA applications cannot be granted and to be closed.

ISTS Expansion in Northern Region

1. Augmentation of ICT (3rd) at 400/220 kV Amargarh (GIS) substation

It was deliberated that Augmentation of 400/220 kV ICT(3rd) at 400/220kV Amargarh (GIS) S/s was agreed in the 3rd NRPC(TP) held on 19.02.2021 with the time frame of March 2026. Same was taken up for discussion in 5th NCT held on 25.08.2021 and 02.09.2021 wherein considering the timeframe of March, 2026 it was decided to defer the same and may be taken up later for approval.

Recently, JKPTCL vide letter dated 17.02.2022 to CEA informed that the 630 MVA 400/220 kV GIS at Amargarh is directly feeding the 450 MVA, 220/132 kV Zainakote GSS and the 320 MVA, 220/132 kV GSS Delina. The transmission plan (Kashmir region) for 2022-27 envisages addition of 2479 MVA at different voltage levels and stands approved in the CEA meeting held on 19-02-2021. This includes 1130 MVA at 220/33 kV Level, 480 MVA at 220/132 kV level and 869 MVA at 132/33 kV level. The commissioning of the new Grid Stations necessitates provision of new sources or capacity augmentation at 400/220 kV level to enable the trickling down of the benefits of the new Grid stations to the consumers. In the JKPTCL letter, it was also stated that the existing sources of power shall not be able to cater to the increased demand of power and this necessitates immediate steps to augment the sources. It is proposed to augment the existing 220/132 kV GSS Zainakote from 450 MVA to 700 MVA and 220/132 kV GSS Delina from 320 MVA to 480 MVA by the end of 2023. The maximum drawl of power from 630

MVA Amargarh (GIS) during the present winter has been about 550 MW. In order to address the growing demand of power, it is imperative that Amargarh (GIS) S/s is augmented by atleast 50% i.e. 315 MVA.

Further, in the Quarterly operational feedback of January 2023, Grid-India has also highlighted the issue of 'N-1' non-compliance of loading of ICTs at Amargarh S/s. In the report it was informed that with 2 no. of 315 MVA ICTs, loading of ICTs was more than 'N-1' compliance limit (>425MW) most of the time in 2022-23 (Quarter 3: Oct'22-Dec'22) during high demand of J&K. Grid-India recommended that new ICT augmentation at Amargarh (GIS) may be expedited.

In the 202nd OCC meeting held on 16.12.22, NRLDC stated that "loading of 400/220kV Amargarh ICTs was above N-1 contingency limits for last 30 days. It was mentioned that proposal for capacity augmentation was discussed in OCC/ NRPC meeting but could not be finalised. Therefore, till capacity is augmented at 400/220kV Amargarh, any N-1 contingency is likely to lead to tripping of both ICTs as they are loaded beyond their N-1 contingency limit and there would be load loss in valley area". Subsequently in the recent 62nd NRPC meeting held on 31.01.22, agenda for N-1 violation related issues in Himalayan states of NR during winter 2022-23 was deliberated. In the meeting, CTU stated that new ICT at 400/220kV Amargarh has already been approved and implementation of these ICTs before their timeline of 2026 would be discussed with CEA.

The loading pattern of Amargarh ICTs of past one year is as below :

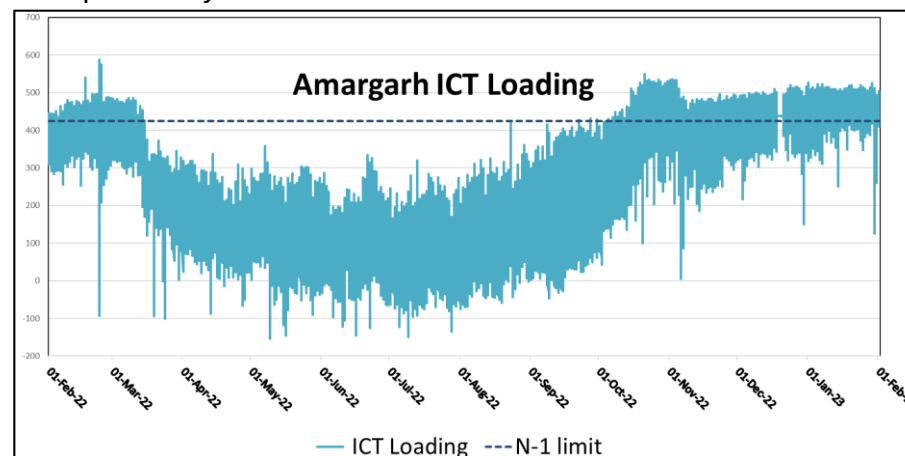


Figure 1: Loading of Amargarh ICTs of Past one year(Source: Grid-India)

From the loading pattern of Amargarh ICTs (2x315 MVA), it is observed that the ICT loading is more than 425MW in the month of October to March and may become N-1 non-compliant.

The S/s owner i.e. M/s INDIGRID confirmed that the space is available for the installation of 400/220kV, 315MVA ICT (3rd) (Three single phase units) at Amargarh SS, however space is not available for 500MVA ICT. Further, transportation of 500 MVA (single phase units) ICT through enroute tunnels may also pose challenge. The details regarding the expansion of PEB Building, GIS BUS -1 & 2, GIS Duct for the connection with ICT, Land levelling need to be explored / identified and same would be provided shortly. Subsequently, M/s IndiGrid vide mail 15.03.22 informed that for installation of 400/220kV, 315MVA ICT (3rd) (Three single phase units) at Amargarh SS, required new Diameter at both sides along with GIB (420kV & 245Kv 1-ph indoor and outdoor GIB). Additionally, it required tertiary bus bar & neutral bus bar extension for the delta & neutral formation for which detail will be provided later.

JKPTCL reiterated the urgent requirement of the ICT augmentation. JKPTCL also stated that in view of increasing demand of J&K, there may be requirement of additional 400/220kV ICT (4th) at Amargarh S/s in future. M/s INDIGRID opined that there is space constraint in existing switchyard to install 4th ICT at Amargarh S/s, however they will explore to acquire additional land nearby to existing Amargarh S/s to accommodate 4th 400/220kV ICT.

M/s INDIGRID stated that implementation timeframe of 21 months is a bit Compressed timeline due to issues in manufacturing as well as transportation of above ICT. INDIGRID requested to increase the time line for implementation of ICT. In reply, it was stated that timeline for implementation of ICT at GIS S/s has been kept as 21 months whereas in AIS S/s it is 18 months. Additionally, 3rd ICT is required on urgent basis due to critical loading in 'N-1' contingency and based on request of JKPTCL. In view of that it is expected that 3rd ICT would be available by Dec'24 (considering 21 months from expected approval in Mar'23), however efforts to be made for implementation of ICT in 18 months. JKPTCL also agreed for the same

Considering the growing demand of power in J&K, as well as to meet the N-1 criteria of loading of ICTs, it is proposed to implement one no. of 400/220kV, 315MVA/500 MVA ICT (3rd) at Amargarh S/s under ISTS. GRID-INDIA stated that 315MVA ICT is preferred over 500MVA ICT as spare (single phase unit of 105MVA) is already available at Amargarh S/s and that could be utilized with proposed 315MVA ICT also. CEA also concurred on installation of 400/220kV, 315MVA ICT (3rd) (Three single phase units) at Amargarh S/s.

In view of that following scope was agreed in ISTS :

- Augmentation of 400/220 kV, 315MVA ICT (single phase units of 105MVA) at Amargarh (GIS) S/s along with associated transformer bays

Implementation Timeframe: 21 months (18 months on best effort basis)

2. Requirement of one set of Bus Sectionalizer at 400kV level of 765/400/220kV Fatehgarh-III PS (Section-2)

It was deliberated that Transmission system for additional 20 GW REZ in Northern Region (Phase-III) including Fatehgarh-III PS (new section) was agreed in the 3rd NRPC (TP) meeting held on 19.02.2021 and 49th NRPC meeting held on 27.09.2021. Subsequently in the 5th NCT meeting held on 25.08.2021 & 02.09.2021, above scheme was agreed for implementation. As part of above scheme, STATCOM along with MSC & MSR at Fatehgarh-III PS was also discussed and it was decided that its requirement would be reviewed at a later date.

Subsequently in the 11th NCT meeting held on 28.12.22 and 17.01.23, It was agreed that $\pm 2 \times 300$ MVar STATCOM along with 4×125 MVar MSC, 2×125 MVar MSR along with 2 Nos. of 400 kV bays at Fatehgarh-3 PS will be implemented as part of transmission schemes "Transmission system for evacuation of power from REZ in Rajasthan (20 GW) Phase-III Part F" which is currently in bidding stage.

It was stated that 765/400/220kV Fatehgarh-III PS (new section) is being implemented by POWERGRID as part of Phase-III Part E1 scheme. In the 11th NCT meeting, it was also noted that ± 300 MVar STATCOM should be placed in each 400 kV section of Fatehgarh-III PS (Phase-III Part E1). Therefore One set of sectionalizer at 400kV level of 765/400/220kV Fatehgarh-III PS (Section-2) is required so as to place ± 300 MVar STATCOM units along with MSC & MSR in each 400kV section of Fatehgarh-III PS (Phase-III Part E1). POWERGRID agreed for the proposal for requirement of above Bus sectionalizer.

In view of above following was agreed as ISTS scheme

- 1 Set of Bus sectionalizer at 400kV level of 765/400/220kV Fatehgarh-III PS (Section-2)

Implementation Timeframe: Matching with implementation of Ph-III Part-E1 package

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Annexure-A

The following provisions shall be applied with respect to grant of connectivity to **RE projects** to the ISTS Grid:

- i. The grant of Stage-I Connectivity shall not create any vested right in favor of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the quarterly progress of development of their generation project and associated transmission infrastructure/dedicated line as per format **RCON-I-M** by 30th day of June and 31st day of December of each year. Further, Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I grantee and their Application fees shall be forfeited.
- ii. If the capacity of the Stage-I Connectivity location is allocated to other Stage-II grantees, the balance Stage-I grantees shall be allocated Stage-II Connectivity to an alternate location.
- iii. For optimization of ROW, it is proposed that transmission towers of various dedicated lines, up to 2-3 km periphery from the entry of the pooling station may be of **Multi-circuit type**. The generation developers/applicants associated with the various pooling stations may coordinate amongst themselves for implementation of such span/stretch of the dedicated connectivity lines with M/C towers.
- iv. The grant of Stage-I / Stage-II Connectivity shall not entitle an applicant to interchange any power with the grid unless it obtains LTA/MTOA/STOA for power transfer requirements. As grant of LTA may require transmission system strengthening, the applicants are advised to apply for LTA immediately, to enable timely transfer of power to the beneficiaries. Also as per the prevailing Transmission Planning Criteria of CEA, "N-1" contingency criteria may not be applied to the immediate connectivity of wind/solar farms with the ISTS/Intra State grid.
- v. Stage-II Connectivity Grantee shall sign the Transmission Agreement for Connectivity and submit the Connectivity Bank Guarantee (**Conn-BG1 and Conn-BG2**) to CTU within 30 days of issue of intimation. No extension of time shall be granted and in case of failure to sign the Agreement and / or to furnish the requisite bank guarantee, Stage-II Connectivity shall be cancelled under intimation to the grantee without any prior notice.
- vi. Two or more Applicants may apply for Stage-II Connectivity at a common bay along with an agreement duly signed between such Applicants for sharing the Dedicated Transmission Line. The Stage-II Connectivity shall be granted to such Applicants subject to availability of capacity in the Dedicated Transmission Line. In such cases, **Conn-BG1 and Conn-BG2**, as applicable as per Clause 10.8 of the Revised RE Procedure, shall be submitted by each such Applicant.
- vii. The Stage-II Connectivity grantee shall furnish progress of the monitoring parameters on quarterly basis in the format given at **FORMAT-RCON-II-M** by the last day of each quarter. Failure to update progress of the monitoring parameters shall be considered as adverse progress and in such case CTU shall approach the Commission for appropriate directions. Further,

- the Stage-II Connectivity grantees shall be required to complete the dedicated transmission line(s) and pooling sub-station(s) as defined under para 11.2(A) of the Revised Detailed Procedure for RE. If the grantee fails to complete the dedicated transmission line within the stipulated period, the Conn-BG1 & Conn-BG2 of the grantee shall be encashed and Stage-II connectivity shall be revoked.
- viii. Applicants after grant of Stage-II Connectivity to the grid shall have to furnish additional details to CTU for signing of “Connection Agreement” as per format given at **FORMAT-CON-4**. The finalized template of generation data for Solar and Wind based generation projects has been appended as additional sections at Para F (Details of Connection – Solar PV Station) and Para G (Details of Connection – Wind Generating Station) in the existing FORMAT-CON-4. The modified FORMAT CON-4 is available on our website (www.powergridindia.com >> CTU Open Access). The Applicants are advised to furnish such details as early as possible for enabling them have lead time for any type of access.
- ix. The CTU will process the above information and will intimate the Connection details as per format given at **FORMAT-CON-5**. Pursuant to such Connection details, the applicant shall have to sign “Connection Agreement” with CTU prior to the physical inter-connection as per format given at **FORMAT-CON-6**. In case the connectivity is granted to the ISTS of an inter-State transmission licensee other than the CTU, a tripartite agreement shall be signed between the applicant, the Central Transmission Utility and such inter-State transmission licensee, in line with the provisions of the Regulations.
- x. Applicants are required to submit the test reports supported vide undertaking as well as compliance certificate from manufacturer for all applicable provisions under the CEA (Technical Standards for Connectivity to the Grid) Regulations, 2007 (as amended) (including the provisions of LVRT/HVRT, active power injection control, dynamically varying reactive power support, limits for Harmonic & DC current injection, Flicker limits etc.) from labs accredited by Govt./NABL/other recognized agencies. In case any discrepancies / incompleteness are found in the documents / test reports submitted to CTU, the connection offer (CON-5) / connection agreement (CON-6) shall not be processed further.
- xi. CEA vide order dated 09.11.2020, has mandated all power generating stations of 0.5 MW or above capacity to register themselves on CEA e-portal and get a **Unique Registration Number (URN)**. The same is required as per Regulation 11 of Technical Standards for Connectivity with the Grid Regulations, 2007 prior to physical inter-connection of the Generating station with ISTS Grid.

Transmission system for LTA (400 MW) in NR

A. Transmission system for present LTA (Appl. No. SW9990894382-M029 D001 A007-1671216520154) at Bhadla-3 PS

- 1) 1x500 MVA (1st), 400/220 kV pooling station at Bhadla-3 PS

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under Phase-III (20 GW) scheme)

- 1) Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-3 substation
- 2) Establishment of 6000 MW, $\pm$ 800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP)
- 3) Bhadla-3 – Bhadla (HVDC) 400 kV D/c (1st) quad moose line along with the line bays at both substations
- 4) Fatehgarh-2 – Bhadla-3 400kV D/c line (Quad) along with 50 MVar Switchable line reactor for each circuit at both ends of Fatehgarh 2- Bhadla-3 400kV D/c line
- 5) Establishment of 2x1500 MVA, 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor
- 6) LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur
- 7) Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation

Transmission system for LTA (100MW) in NR

A. Transmission system for present LTA (Appl. No. 0412100012 & 0412100013) at Bhadla-III PS

- 1) 1x500 MVA (1st) 400/220 kV ICT at Bhadla-III PS

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under Phase-III (20 GW) scheme)

- 1) Establishment of 6000 MW, $\pm$ 800 kV Bhadla (HVDC) [LCC] terminal station (4x1500 MW) at a suitable location near Bhadla-3 substation
- 2) Establishment of 6000 MW, $\pm$ 800 kV Fatehpur (HVDC) [LCC] terminal station (4x1500 MW) at suitable location near Fatehpur (UP)
- 3) Bhadla-3 – Bhadla (HVDC) 400 kV D/c (1st) quad moose line along with the line bays at both substations
- 4) Fatehgarh-2 – Bhadla-3 400kV D/c line (Quad) along with 50 MVar Switchable line reactor for each circuit at both ends of Fatehgarh 2- Bhadla-3 400kV D/c line
- 5) Establishment of 2x1500 MVA, 765/400 kV ICTs at Fatehpur (HVDC) along with 2x330 MVar (765kV) bus reactor
- 6) LILO of both ckts of 765 kV Varanasi – Kanpur (GIS) D/c line at Fatehpur
- 7) Augmentation by 1x1500 MVA ICT at 765/400kV Kanpur (GIS) substation

Transmission system for LTA (0412100018: 50 MW) at Bhadla-II PS**A. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)**

1. Augmentation of 2x1500 MVA, 765/400 kV ICT (3rd & 4th) at Bhadla-II Pooling station
2. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVAR at 400kV level & 2x330 MVAR bus reactors at 765kV level at Sikar -II
3. Bhadla-II PS – Sikar-II 765kV D/c line (2nd)
4. Sikar-II – Neemrana 400kV D/c line (Twin HTLS)*
5. 1x330 MVAR Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
6. 1x240 MVAR Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
7. Sikar-II – Aligarh 765kV D/c line
8. 1x330 MVAR switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line
9. Establishment of 765/400 kV, 3X1500 MVA GIS substation at Narela with 765 kV (2x330 MVAR) bus reactor and 400kV (1x125 MVAR) bus reactor.
10. Khetri – Narela 765 kV D/c line
11. LILO of 765 kV Meerut- Bhiwani S/c line at Narela
12. 1x330 MVAR Switchable line reactor for each circuit at Narela end of Khetri – Narela 765kV D/c line
13. Removal of LILO of Bawana – Mandola 400kV D/c (Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharani Bagh – Narela 400kV D/c(Quad) and Maharani Bagh-Gopalpur-Narela 400 kV D/c (Quad) lines.
14. 2 no of line bays at Narela each for Maharani Bagh – Narela 400 kV D/c (Quad) and Maharani Bagh –Gopalpur**-Narela 400 kV D/c (Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s.

*with minimum capacity of 2100 MVA on each circuit at nominal voltage

Additional Scheme to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose)

1. Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
2. Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km

with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.*
with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipments each at Jodhpur (Surpura)(RVPN) and Kankroli S/s(3150 A)

Transmission system for LTA (120 MW) at Fatehgarh-IV PS

A. Transmission system for present LTA (Appl. No. 0412100019) at Fatehgarh-IV PS (Section-1)

- 1) Augmentation of 1x500 MVA (2rd), 400/220kV ICT at Fatehgarh-IV pooling station (Section-1)

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under Phase-III (20 GW) scheme)

- 1) Fatehgarh-IV- Fatehgarh-III 400 kV D/c (Twin HLTS*) line
- 2) Augmentation of 2x1500 MVA (4th and 5th), 765/400kV ICT at Fatehgarh-3 pooling station (Section-II)
- 3) Establishment of 2x1500MVA 765/400kV Substation at suitable location near Beawar along with 2x330 MVar, 765kV Bus Reactor & 2x125 MVar, 420kV bus Reactor
- 4) Fatehgarh-3– Beawar 765 kV D/c (2nd) along with 330 MVar Switchable line reactor for each circuit at each end of Fatehgarh-3– Beawar 765 kV D/c line (2nd)
- 5) Fatehgarh 3- Bhadla-3 400kV D/c line (Quad) along with 50 MVar Switchable line reactor for each circuit at both ends of Fatehgarh 3-Bhadla-3 400kV D/c line
- 6) LILO of both circuit of Ajmer-Chittorgarh 765 kV D/c line at Beawar substation
- 7) LILO of 400kV Kota –Merta line at Beawar substation
- 8) Establishment of 2x1500 MVA 765/400kV substation at suitable location near Dausa along with 2x330 MVar, 765 kV Bus Reactor & 2x125 MVar, 420 kV bus Reactor
- 9) LILO of both circuits of Jaipur (Phagi)- Gwalior 765 kV 2xS/c at Dausa along with 240 MVar Switchable line reactor for each circuit at Dausa end of Dausa – Gwalior 765 kV D/c line
- 10) LILO of both circuits of Agra – Jaipur(south) 400kV D/c at Dausa along with 50 MVar Switchable line reactor for each circuit at Dausa end of Dausa – Agra 400kV D/c line
- 11) Beawar – Dausa 765 kV D/c line along with 240 MVar Switchable line reactor for each circuit at each end
- 12) STATCOM:
Fatehgarh – III S/s: STATCOM: $\pm$ 2x300 MVar, 4x125 MVar MSC, 2x125 MVar MSR

Transmission system for LTA (200 MW) at Fatehgarh-III PS (Section-II)

A. Transmission system for present LTA (Appl. No. 0412100020) at Fatehgarh-III PS (Section-II)

- 1) Augmentation of 1x500 MVA (3rd), 400/220kV ICT at Fatehgarh-3 pooling station (Section-II)

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under Phase-III (20 GW) scheme)

- 1) Augmentation of 2x1500 MVA (4th and 5th), 765/400kV ICT at Fatehgarh-3 pooling station (Section-II)
- 2) Establishment of 2x1500MVA 765/400kV Substation at suitable location near Beawar along with 2x330 MVar, 765kV Bus Reactor & 2x125 MVar, 420kV bus Reactor
- 3) Fatehgarh-3– Beawar 765 kV D/c (2nd) along with 330 MVar Switchable line reactor for each circuit at each end of Fatehgarh-3– Beawar 765 kV D/c line (2nd)
- 4) Fatehgarh 3- Bhadla-3 400kV D/c line (Quad) along with 50 MVar Switchable line reactor for each circuit at both ends of Fatehgarh 3- Bhadla-3 400kV D/c line
- 5) LILO of both circuit of Ajmer-Chittorgarh 765 kV D/c line at Beawar substation
- 6) LILO of 400kV Kota –Merta line at Beawar substation
- 7) Establishment of 2x1500 MVA 765/400kV substation at suitable location near Dausa along with 2x330 MVar, 765 kV Bus Reactor & 2x125 MVar, 420 kV bus Reactor
- 8) LILO of both circuits of Jaipur (Phagi)- Gwalior 765 kV 2xS/c at Dausa along with 240 MVar Switchable line reactor for each circuit at Dausa end of Dausa – Gwalior 765 kV D/c line
- 9) LILO of both circuits of Agra – Jaipur(south) 400kV D/c at Dausa along with 50 MVar Switchable line reactor for each circuit at Dausa end of Dausa – Agra 400kV D/c line
- 10) Beawar – Dausa 765 kV D/c line along with 240 MVar Switchable line reactor for each circuit at each end
- 11) STATCOM:
Fatehgarh – III S/s: STATCOM: $\pm$ 2x300 MVar, 4x125 MVar MSC, 2x125 MVar MSR

Additional Scheme to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose)

- 1) Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
- 2) Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km
with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.*
with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipments each at Jodhpur (Surpura)(RVPN) and Kankroli S/s(3150 A)

Transmission system for LTA (100 MW) at Fatehgarh-III PS (Section-I)

A. Transmission system for present LTA (Appl. No. 0412100021) at Fatehgarh-III PS (Section-I)

1. Establishment of 2x500 MVA (3rd & 4th) 400/220kV ICTs at Fatehgarh-III Pooling station

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

1. Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line
2. Jaisalmer (RVPN) S/s - Fatehgarh-III PS (Twin HTLS) D/c line
3. Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
4. Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
5. 1x240 MVar Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
6. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVar at 400kV level & 2x330 MVar bus reactors at 765kV level at Sikar -II
7. Sikar-II – Neemrana 400kV D/c line (Twin HTLS)
8. Bhadla-II PS – Sikar-II 765kV D/c line
9. 1x330 MVar Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
10. 1x240 MVar Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
11. Sikar-II – Aligarh 765kV D/c line
12. 1x330 MVar switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line

Transmission system for LTA (300 MW) at Fatehgarh-III PS (Section-I)

A. Transmission system for present LTA (App No. (0412100022) at Fatehgarh-III PS (Section-I)

1. Establishment of 2x500 MVA (3rd & 4th) 400/220kV ICTs at Fatehgarh-III Pooling station

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

1. Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line
2. Jaisalmer (RVPN) S/s - Fatehgarh-III PS (Twin HTLS) D/c line
3. Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
4. Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
5. 1x240 MVar Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
6. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVar at 400kV level & 2x330 MVar bus reactors at 765kV level at Sikar -II
7. Sikar-II – Neemrana 400kV D/c line (Twin HTLS)
8. Bhadla-II PS – Sikar-II 765kV D/c line
9. 1x330 MVar Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
10. 1x240 MVar Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
11. Sikar-II – Aligarh 765kV D/c line
12. 1x330 MVar switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line

Transmission system for LTA (100 MW) at Fatehgarh-III PS (Section-I)**Transmission system for present LTA (App No. 0412100023) at Fatehgarh-III PS (Section-I)**

1. Augmentation of 1x500 MVA (4th) 400/220kV ICT at Fatehgarh-III Pooling station

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

1. Fatehgarh-II PS – Fatehgarh-III PS 400 kV (Twin HTLS) D/c line
2. Jaisalmer (RVPN) S/s - Fatehgarh-III PS (Twin HTLS) D/c line
3. Augmentation with 765/400kV, 2x1500MVA Transformer (5th & 6th) at Fatehgarh-II PS
4. Fatehgarh-II PS – Bhadla-II PS 765kV D/c line (2nd)
5. 1x240 MVar Switchable line reactor for each circuit at each end of Fatehgarh-II – Bhadla-II 765kV D/c line
6. Establishment of 765/400kV, 2x1500 MVA S/s at suitable location near Sikar (Sikar-II Substation) with 1x125 MVar at 400kV level &
7. 2x330 MVar bus reactors at 765kV level at Sikar -II
8. Bhadla-II PS – Sikar-II 765kV D/c line
9. Sikar-II – Neemrana 400kV D/c line (Twin HTLS)*
10. 1x330 MVar Switchable line reactor for each circuit at Sikar-II end of Bhadla-II –Sikar-II 765kV D/c line
11. 1x240 MVar Switchable line reactor for each circuit at Bhadla-II end of Bhadla-II – Sikar-II 765kV D/c line
12. Sikar-II – Aligarh 765kV D/c line
13. 1x330 MVar switchable line reactor for each circuit at each end of Sikar-II – Aligarh 765kV D/c line
14. Establishment of 765/400 kV, 3X1500 MVA GIS substation at Narela with 765 kV (2x330 MVar) bus reactor and 400kV (1x125 MVar) bus reactor.
15. Khetri – Narela 765 kV D/c line
16. LILO of 765 kV Meerut- Bhiwani S/c line at Narela
17. 1x330 MVar Switchable line reactor for each circuit at Narela end of Khetri – Narela 765kV D/c line
18. Removal of LILO of Bawana – Mandola 400kV D/c (Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from
19. Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharaniabagh – Narela 400kV D/c(Quad) and Maharaniabagh-Gopalpur-Narela 400 kV D/c (Quad) lines.
20. 2 no of line bays at Narela each for Maharaniabagh – Narela 400 kV D/c (Quad) and Maharaniabagh –Gopalpur-Narela 400 kV D/c (Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s.
21. STATCOM:
Fatehgarh – II S/s: STATCOM: ± 600 MVar, 4x125 MVAR MSC, 2x125 MVar MSR
Bhadla – II S/s: STATCOM: ± 600 MVar, 4x125 MVAR MSC, 2x125 MVar MSR
*with minimum capacity of 2100 MVA on each circuit at nominal voltage

List of Participants of 16th Consultation meeting for Evolving Transmission Schemes in NR held on 28.02.2023

CEA

Shri Kanhaiya Singh Kushwaha Assistant Director

SECI

Shri R.K.Agarwal Consultant

GRID INDIA

Shri Gaurav Malviya Manager

POWERGRID

Shri Somiran Das GM

CTU

Shri Kashish Bhambhani GM (CTU)

Shri Laxmi Kant Sr. DGM (CTU)

Shri Sandeep Kumawat Ch. Manager (CTU)

Smt. Ankita Singh Ch. Manager (CTU)

Shri R Narendra Sathvik Manager (CTU)

Shri Chinmay Sharma Manager (CTU)

Shri Roushan Kumar Engineer (CTU)

Smt Namrata Singh Engineer (CTU)

Shri Madhusudan Meena Engineer (CTU)

HVPNL

Shri Rajesh Kumar Jangra Xen System Study

PTCUL

Shri Saima Kamal	Superintending Engineer
Shri Lalit Kumar	SE
Shri Ashok Kumar	EE

PSTCL

Shri Nitin Kumar	Sr. XEN/Planning-1
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JKPTCL

Shri Ehtisham Andrabi	Assistant Executive Engineer
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DTL

Shri Dinesh Singh	Mgr. (T)SS & LM. Plg
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LTA/Connectivity Applicants

Shri Abhilash Yadav	AMP Energy Green Private Limited
Shri Ankur Sharma	AMP Energy Green Private Limited
Shri Shashank Gupta	Azure Power India Private Limited
Shri Rajesh Kumar	Azure Power India Private Limited
Shri Ayush Prasad	Amplus Ages Private Limited
Shri Mahendra Singh Dabi	Adani Solar Energy AP Three Limited
Shri Rajesh Kumar Gupta	Adani Solar Energy AP Three Limited
Shri Jyotiprakash Agarwal	Cannice Renewables Energy Private Limited
Shri Lalit Mogha	Cannice Renewables Energy Private Limited
Smt Priyanka Chaurasia	Deshraj Solar Energy Private Limited
Shri Shubham Roy	Deshraj Solar Energy Private Limited

Shri Kumar Vipul	Helia Energy Park Private Limited
Shri Lokendra Singh Ranawat	IndiGrid
Shri Vivek Karthikeyan	IndiGrid
Shri Naman Agarwal	IndiGrid
Shri Pavan Kumar Gupta	Juniper Green Beta Private Limited & Juniper Green Stellar Private Limited
Shri Saurabh Gupta	Khaba Renewable Energy Private Limited
Shri Atul Bachikar	Luceo Solar Private Limited
Shri Sandeep Jain	Prerak Greentech Solar Private Limited
Shri Ajay	Prerak Greentech Solar Private Limited
Shri Kumar Vipul	Radiant Star Solar Park Private Limited
Shri Mohit Jain	ReNew Surya Vihaan, ReNew Surya Aayan & ReNew Surya Roshni Private Limited
Shri Rajas Ranjan Acharya	Resco Global Wind Services Private Limited
Shri Arzaan Dordi	Serentica Renewables India Private Limited
Shri Jaideep tak	Sprng Nirjara Energy Private Limited
Shri Tushar Kapuria	Saimaa Solar Private Limited
Shri Radheshyam Goyal	Tepsol Sun Sparkle Private Limited
Shri Sandeep Anand	Tepsol Sun Sparkle Private Limited
Shri K A Vishwanath	XL Xergi Power Private Limited
Shri Dharmendra Gupta	XL Xergi Power Private Limited